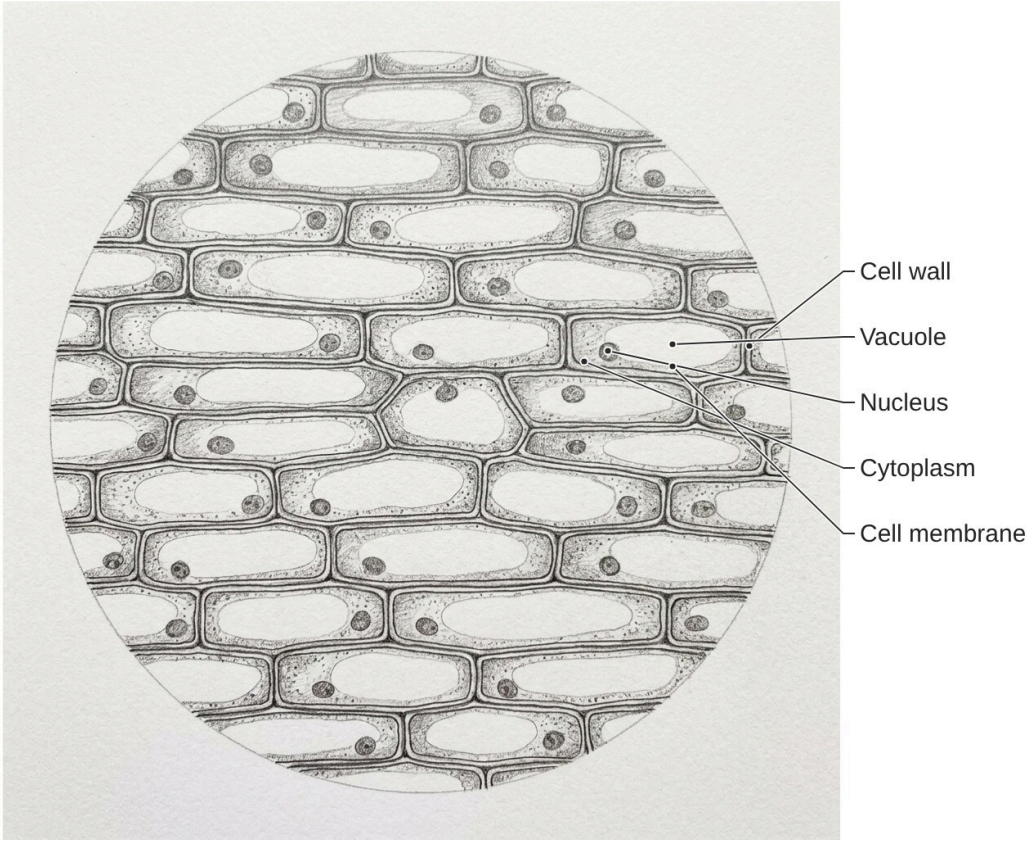
BIO 101L Lab Experiments 2–4

Oct 26, 2026 · @ABEDIN

Experiment 2: Observation of plant cells (onion peel) under a compound microscope

Name of the experiment: Study and observation of plant cells (onion peel) under a compound light microscope

Diagram (plain page): Draw the cells in pencil, label cell wall, cell membrane, cytoplasm, nucleus and vacuole, and write **Onion Peel Cells (Plant Cells)** under the drawing.



Onion Peel Cells (Plant Cells)

stained with safranin, high power

Objective: To prepare a temporary stained wet mount of onion peel and observe plant cells under a compound microscope.

Materials required: Onion, forceps (tweezers), glass slide, cover slip, needle or pin, dropper, water, safranin (or iodine) solution, blotting paper, compound microscope.

Introduction: An onion bulb is made of fleshy layers, each separated by a thin, transparent skin (epidermis) that is only one cell thick. Light passes through it easily, so its cells can be seen clearly under a microscope. A stain such as safranin or iodine colours

the cell wall and nucleus, which makes them easier to see.

Procedure:

1. Put a few drops of water on a clean glass slide so the peel does not dry out or wilt.
2. Take a small piece of onion and, with forceps, peel off the thin membrane from its inner (rough) side.
3. Place the membrane flat on the slide, without folds.
4. Add a drop of safranin or iodine solution to the onion skin.
5. Using a pin, lower a cover slip slowly onto the slide so that no air bubbles are trapped. Remove extra liquid with blotting paper.
6. Put the lowest power objective lens in line with the tube, switch on the light and place the slide on the stage.
7. Looking from the side (not through the eyepiece), lower the tube with the coarse focus knob until the objective is just above the cover slip.
8. Now look through the eyepiece and turn only the fine focus knob to raise the tube until the cells come into focus.
9. Change to a higher power objective to see the cells and their nuclei at greater magnification.

Observation:

1. Many cells are seen packed closely together, with no spaces between them. Together they look like lizard skin or bricks in a wall.
2. Each cell is rectangular and has a definite shape.
3. Each cell is surrounded by a thick, distinct cell wall.
4. A darkly stained, round nucleus is seen in each cell, usually near the edge.
5. A large central vacuole fills most of the cell, pushing the cytoplasm to the sides.

Result: The cells have a definite rectangular shape, a cell wall and a large central vacuole. So, the onion peel cells are plant cells.

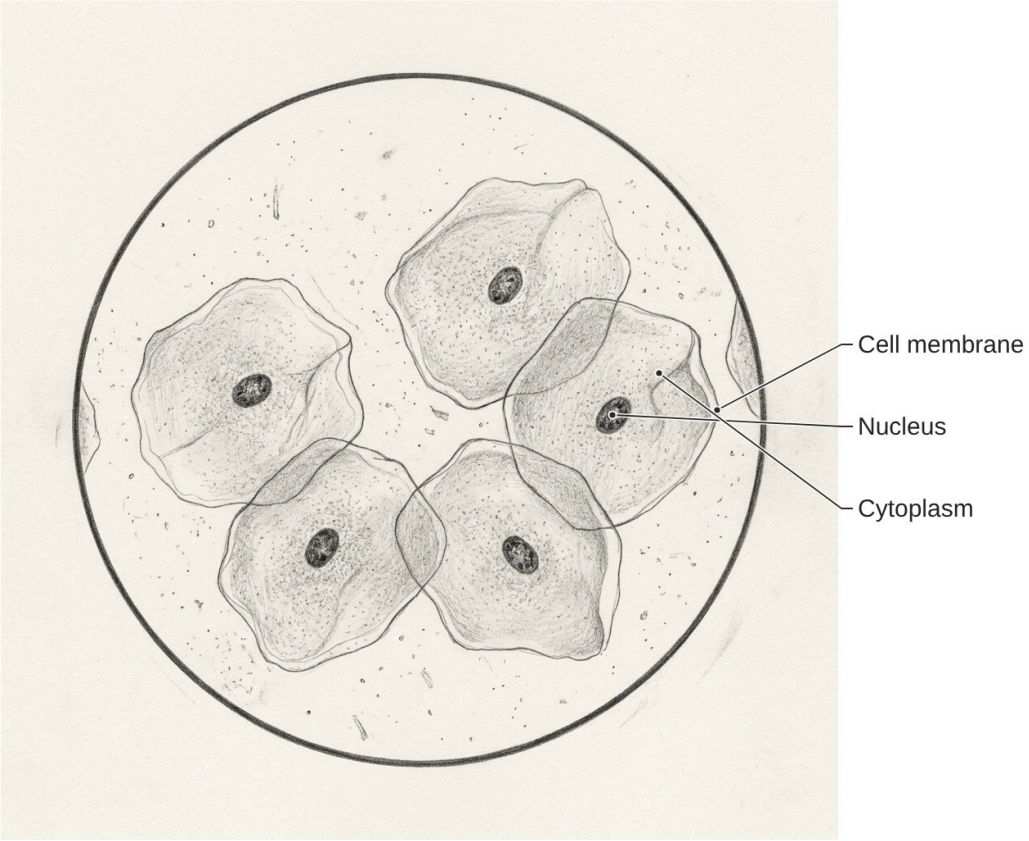
Precautions:

- Peel the membrane carefully so that it is thin and not folded.
- Lower the cover slip slowly with a pin to avoid air bubbles.
- Never use the coarse focus knob while looking through the eyepiece, so the slide and lens are not damaged.
- Always start focusing with the low power objective.
- Handle the stain carefully; it can stain skin and clothes.

Experiment 3: Observation of animal cells (human cheek cells) under a compound microscope

Name of the experiment: Study and observation of animal cells (human cheek cells) under a compound light microscope

Diagram (plain page): Draw the cells in pencil, label cell membrane, cytoplasm and nucleus, and write **Human Cheek Cells (Animal Cells)** under the drawing.



Human Cheek Cells (Animal Cells)

stained with methylene blue, high power

Objective: To prepare a temporary stained mount of human cheek cells and observe animal cells under a compound microscope.

Materials required: Glass slides, cover slips, sterile cotton swabs (individually packed), methylene blue solution (0.5% to 1%: about 1 part stock solution to 4 parts water), plastic pipette or dropper, paper towels or tissue, gloves, compound microscope.

Introduction: The inner lining of the cheek is made of thin, flat epithelial cells that come off easily when gently scraped. Animal cells are almost colourless, so methylene blue is used to stain them. It stains the nucleus deep blue, which makes the cell and its nucleus easy to see.

Procedure:

1. Take a clean cotton swab and gently scrape the inside of your cheek.

2. Smear the swab on the centre of a clean glass slide for 2 to 3 seconds.
3. Add one drop of methylene blue solution on the smear.
4. Place a cover slip on top, avoiding air bubbles.
5. Remove extra solution by touching a paper towel to one side of the cover slip.
6. Place the slide on the microscope stage with the 4x or 10x objective in position and find a cell.
7. Then view the cells at higher magnification.

Observation:

1. A large number of flat, irregular-shaped cells are seen.
2. The cells do not have a cell wall. Each cell is surrounded only by a thin cell membrane.
3. A deeply stained nucleus is seen at the centre of each cell.
4. No prominent vacuoles are seen in the cells.

Result: The cells observed have no cell wall and no prominent vacuole. So, the cells on the slide are animal cells.

Precautions:

- Use a new sterile cotton swab, and scrape the cheek gently.
- Wear gloves when handling methylene blue. Concentrated methylene blue is toxic if swallowed.
- Do not use too much stain; remove the extra with a paper towel.
- Lower the cover slip carefully to avoid air bubbles.
- Dispose of used swabs safely and wash your hands after the experiment.

Experiment 4: Identification and classification of five animal specimens

Name of the experiment: Identification and classification of five animal specimens

Diagram (plain pages): Draw each specimen on its own plain page in pencil, label the parts listed under it, and write its name under the drawing (for example, **Octopus (Octopus vulgaris)**).

Objective: To identify and classify five animal specimens and study their identifying characteristics, habitat and feeding behaviour.

Introduction: Animals are grouped by the features they share, such as body form, symmetry and body covering. Each animal is placed in a series of ranks: kingdom, phylum, class, order, family, genus and species. Every animal gets a two-part scientific name (genus and species), which is the same all over the world.

Materials required: Preserved specimens (octopus, magpie robin, starfish, seahorse, grasshopper), hand lens, pencil, eraser, practical notebook.

Procedure:

1. Observe each specimen carefully, using a hand lens where needed.
2. Note its identifying characteristics.
3. Classify it from kingdom down to species.
4. Record its habitat and feeding behaviour.
5. Draw each specimen neatly in pencil and label the important body parts.

Note: "sp." means the exact species has not been specified. Check the laboratory label to confirm the exact specimen.

Classification:

Rank	Octopus	Magpie robin	Starfish	Seahorse	Grasshopper
Kingdom	Animalia	Animalia	Animalia	Animalia	Animalia
Phylum	Mollusca	Chordata	Echinodermata	Chordata	Arthropoda
Class	Cephalopoda	Aves	Asteroidea	Actinopterygii	Insecta
Order	Octopoda	Passeriformes	Forcipulatida	Syngnathiformes	Orthoptera
Family	Octopodidae	Muscicapidae	Asteriidae	Syngnathidae	Acrididae
Genus	*Octopus*	*Copsychus*	*Asterias*	*Hippocampus*	*Locusta*
Species	*Octopus vulgaris*	*Copsychus saularis*	*Asterias* sp.	*Hippocampus* sp.	*Locusta* sp.

Specimen 1: Octopus (*Octopus vulgaris*)

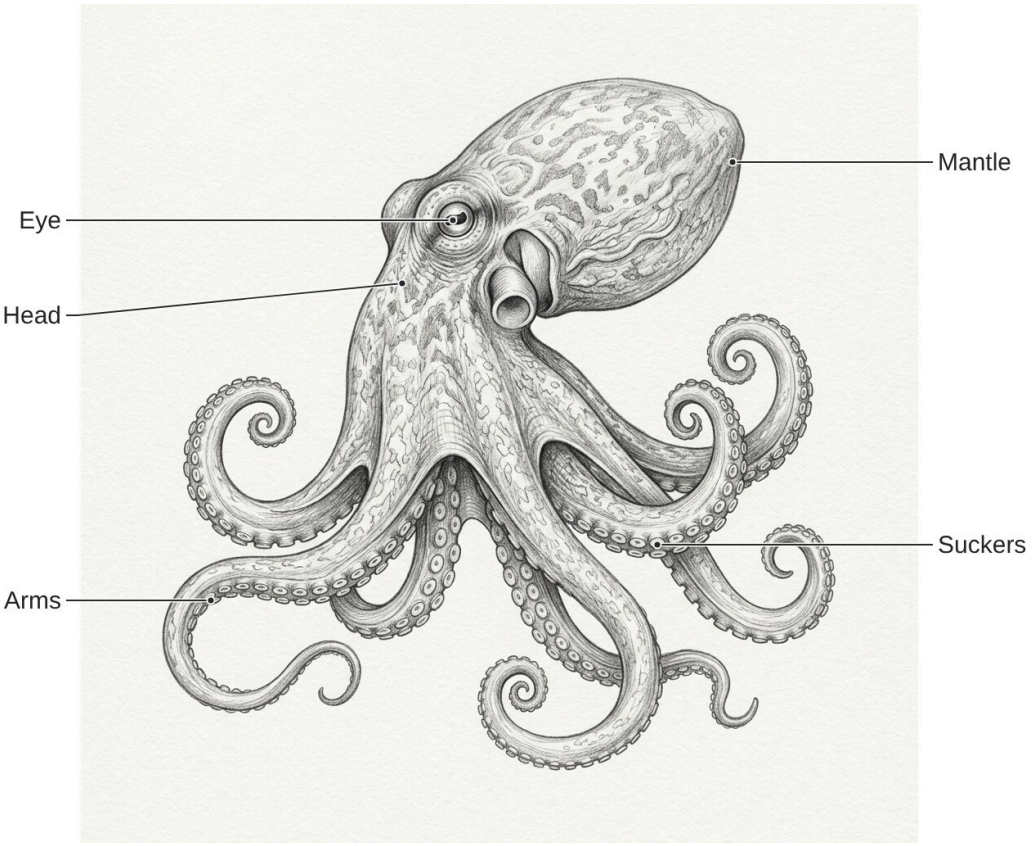
Identifying characteristics:

1. The body is soft and has no external shell.
2. It has eight arms with suckers.
3. It has a well-developed head and eyes.
4. It can change colour for camouflage.

Habitat: Marine water, including rocky seabeds and coral reefs.

Feeding behaviour: Carnivorous; feeds on crabs, shrimps, molluscs and small fish.

Diagram labels: Mantle, head, eyes, arms, suckers.



Octopus (*Octopus vulgaris*)

Specimen 2: Oriental magpie robin, Doyel (*Copsychus saularis*)

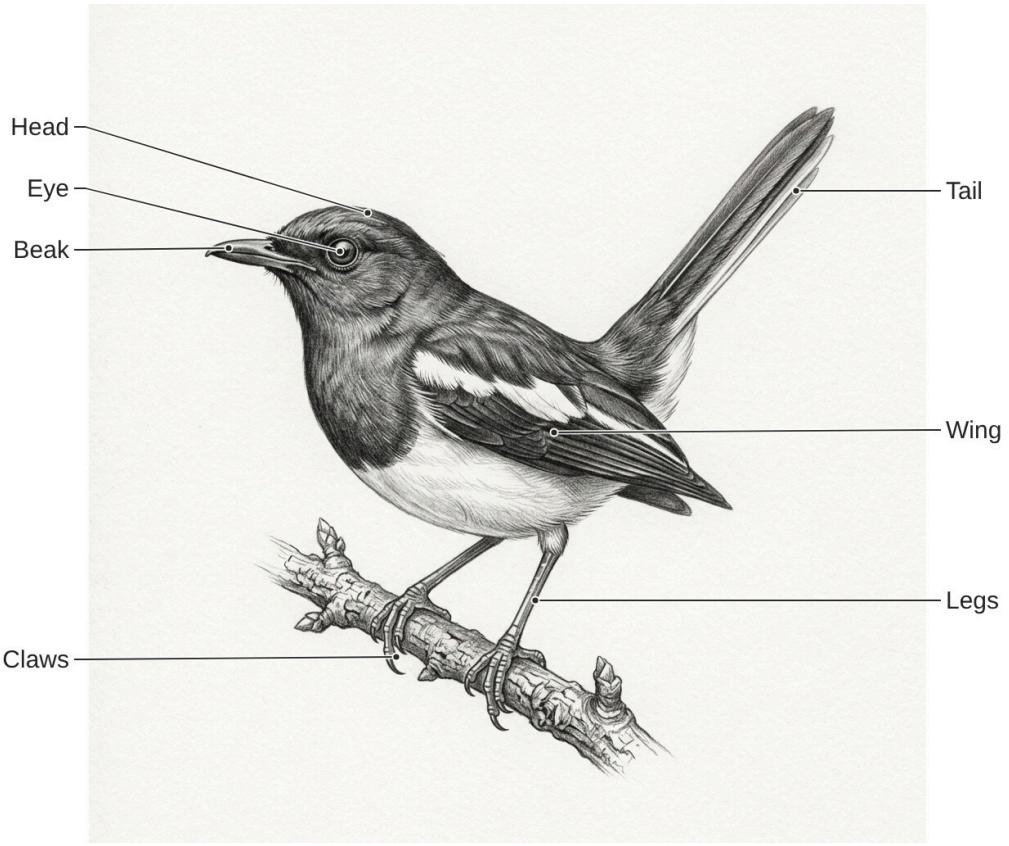
Identifying characteristics:

1. The body is covered with feathers.
2. The male has black-and-white plumage.
3. It has a short, pointed beak.
4. The forelimbs are modified into wings.

Habitat: Gardens, villages, woodland edges and human settlements.

Feeding behaviour: Omnivorous; eats insects, worms and other small invertebrates, and sometimes fruits.

Diagram labels: Beak, eye, head, wing, tail, legs, claws.



Oriental Magpie Robin (*Copsychus saularis*)

Specimen 3: Starfish (*Asterias* sp.)

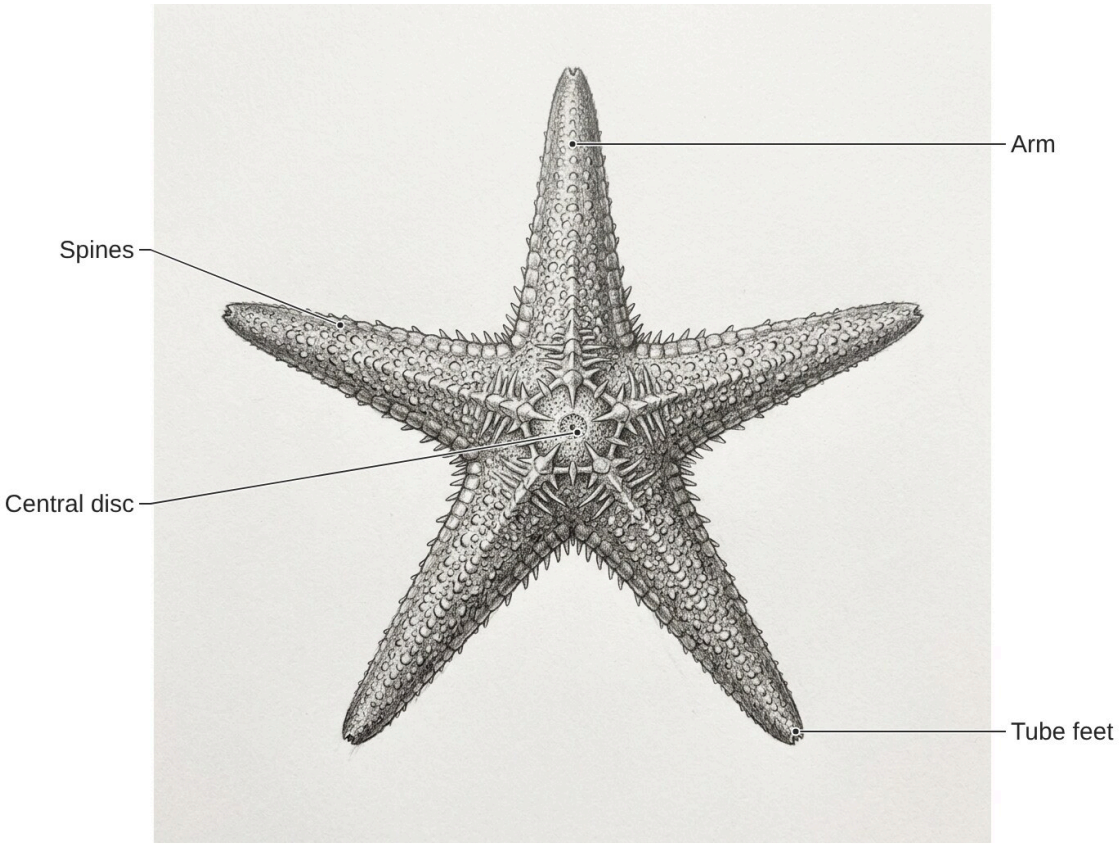
Identifying characteristics:

1. The body is star-shaped, usually with five arms.
2. The skin is rough and spiny.
3. Tube feet are present on the underside.
4. The adult shows radial symmetry.

Habitat: Marine water, usually on the sea bottom.

Feeding behaviour: Carnivorous; feeds on molluscs and other small marine animals.

Diagram labels: Arms, central disc, spines, tube feet.



Starfish (*Asterias* sp.)

upper side; the tube feet are on the underside of each arm

Specimen 4: Seahorse (*Hippocampus* sp.)

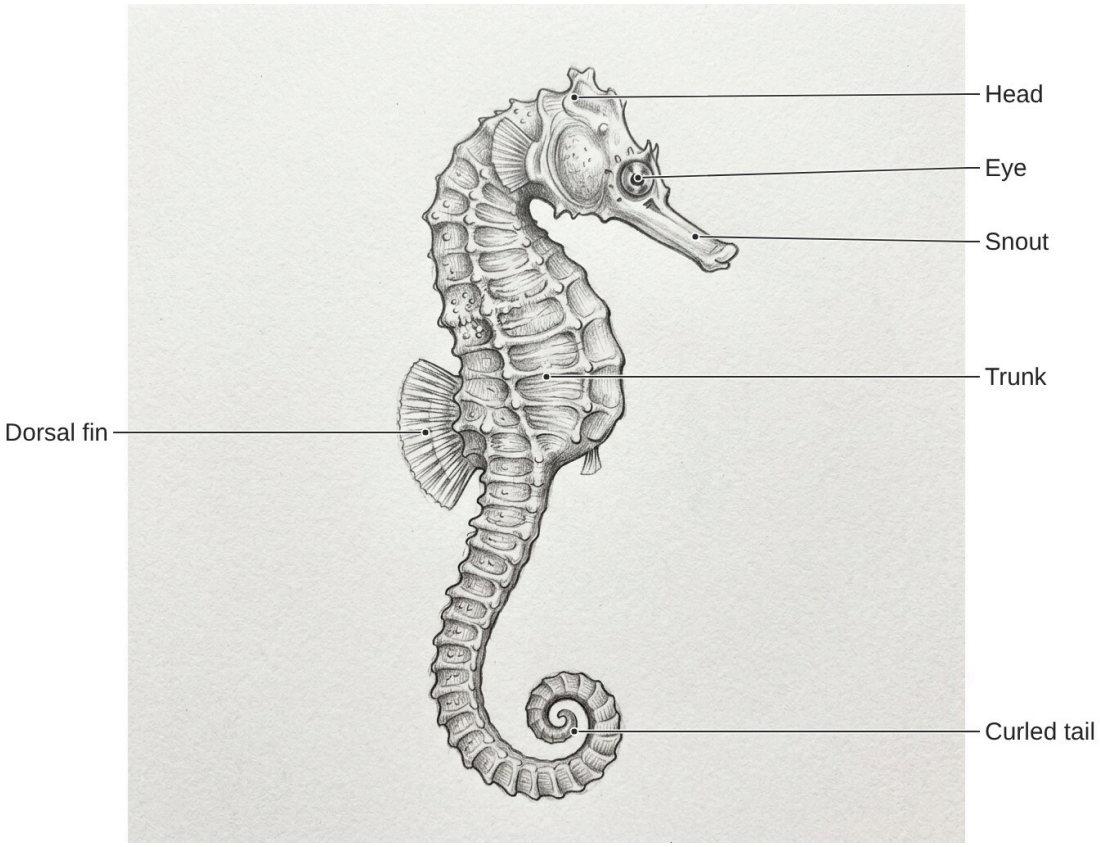
Identifying characteristics:

1. The head looks like that of a horse.
2. The body is covered with bony plates.
3. It has a long, prehensile (grasping) tail.
4. It swims with the help of fins.

Habitat: Shallow marine waters, seagrass beds and coral reefs.

Feeding behaviour: Carnivorous; feeds on small crustaceans and tiny aquatic organisms.

Diagram labels: Head, snout, eye, trunk, dorsal fin, curled tail.



Seahorse (*Hippocampus* sp.)

Specimen 5: Grasshopper (*Locusta* sp.)

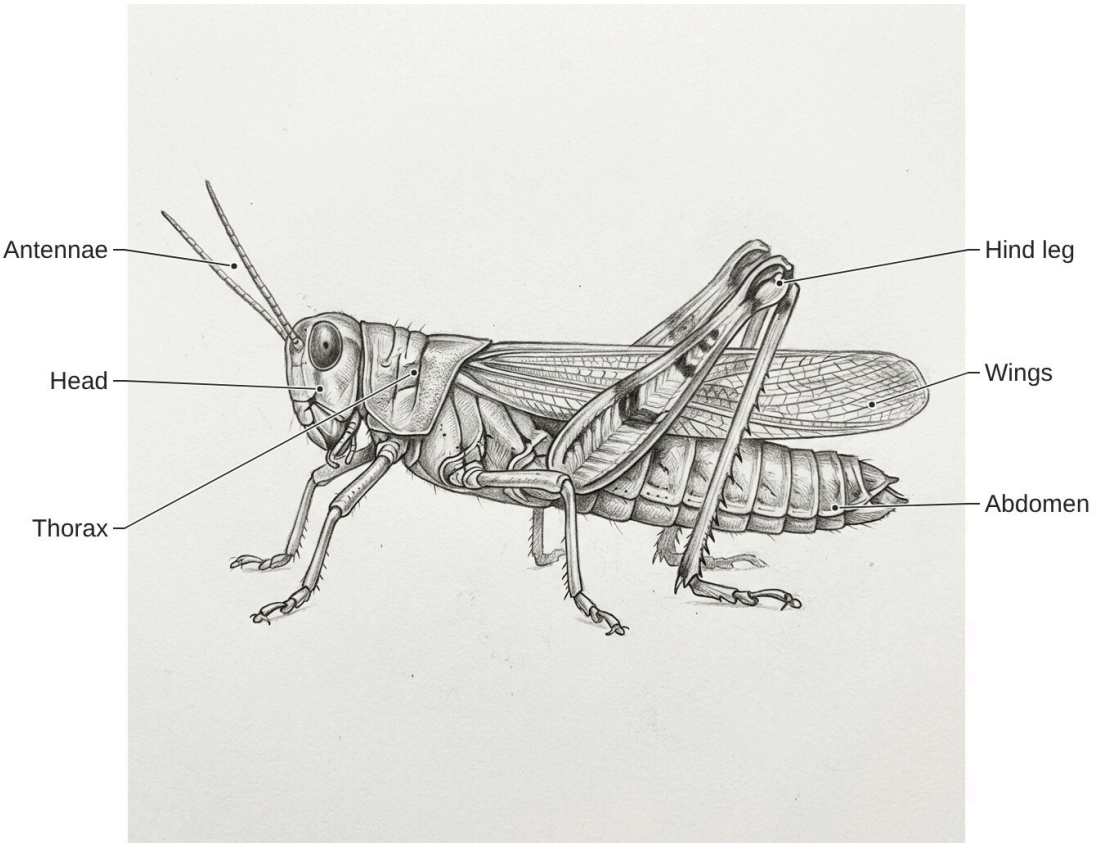
Identifying characteristics:

1. The body is divided into head, thorax and abdomen.
2. It has three pairs of legs.
3. The hind legs are adapted for jumping.
4. It usually has two pairs of wings.

Habitat: Grasslands, fields, gardens and agricultural areas.

Feeding behaviour: Herbivorous; feeds on grass, leaves and other plant parts.

Diagram labels: Antennae, head, thorax, abdomen, wings, hind legs.



Grasshopper (*Locusta* sp.)

Result: The five animal specimens were identified and studied. Their classification, identifying characteristics, habitat and feeding behaviour were recorded.

Precautions:

- Handle the specimens and their jars carefully.
- Do not take preserved specimens out of their jars.
- Avoid touching the preserving liquid; wear gloves if needed.
- Draw in pencil and label the parts neatly.
- Wash your hands after the experiment.